

## 12.6 Bayesian thread II: credible intervals and predictive distributions

Section 7.6 derived the posterior distribution of  $(\boldsymbol{\beta}, \sigma^2)$  under the conjugate normal-inverse-gamma prior, and showed that the flat prior  $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$  reproduces least squares. Here the posterior is summarized by credible regions and used to predict. In the conjugate model these have the shapes of the frequentist sets, and under the flat prior they coincide with them. We end by asking how often a credible interval covers in repeated sampling.

### 12.6.1 Credible sets

#### Definition 12.6.1 — Credible set

Let  $\theta$  be a function of the parameters and let  $\pi(\cdot | \mathbf{y})$  be its posterior distribution. A set  $C(\mathbf{y})$  with  $\Pr\{\theta \in C(\mathbf{y}) | \mathbf{y}\} \geq 1 - \alpha$  is a  $1 - \alpha$  **credible set** for  $\theta$ . If the posterior has a density  $\pi(\theta | \mathbf{y})$ , a set of the form  $\{\theta : \pi(\theta | \mathbf{y}) \geq k\}$  with posterior probability  $1 - \alpha$  is a **highest posterior density (HPD)** set.

Here the statement concerns the realized set. An HPD set has the smallest volume among sets with its posterior probability. We need three facts about the multivariate  $t$  distribution of Definition 7.6.2.

#### Lemma 12.6.1 — Multivariate $t$

Let  $\mathbf{u} \sim t_\nu(\mathbf{m}, \mathbf{S})$  in  $\mathbb{R}^p$ , with  $\mathbf{S}$  positive definite.

- (a) If  $\mathbf{A}$  is  $q \times p$  of rank  $q$ , then  $\mathbf{A}\mathbf{u} \sim t_\nu(\mathbf{A}\mathbf{m}, \mathbf{A}\mathbf{S}\mathbf{A}^\top)$ .
- (b)  $(\mathbf{u} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{u} - \mathbf{m})/p \sim F(p, \nu)$ .
- (c)  $\mathbf{u}$  has density proportional to  $\{1 + (\mathbf{u} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{u} - \mathbf{m})/\nu\}^{-(\nu+p)/2}$ .

*Proof.* By Definition 7.6.2,  $\mathbf{u} = \mathbf{m} + \mathbf{Z}/\sqrt{W/\nu}$  with  $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{S})$  and  $W \sim \chi^2(\nu)$  independent.

- (a)  $\mathbf{A}\mathbf{u} = \mathbf{A}\mathbf{m} + \mathbf{A}\mathbf{Z}/\sqrt{W/\nu}$ , and  $\mathbf{A}\mathbf{Z} \sim N_q(\mathbf{0}, \mathbf{A}\mathbf{S}\mathbf{A}^\top)$  (Theorem 3.3.1) is independent of  $W$ , with  $\mathbf{A}\mathbf{S}\mathbf{A}^\top$  positive definite.
- (b)  $(\mathbf{u} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{u} - \mathbf{m}) = \mathbf{Z}^\top \mathbf{S}^{-1} \mathbf{Z}/(W/\nu)$ , and  $\mathbf{Z}^\top \mathbf{S}^{-1} \mathbf{Z} \sim \chi^2(p)$  independently of  $W$  (Corollary 4.3.3(a)). Divide by  $p$  and use Definition 4.1.2.
- (c) Given  $W = w$ ,  $\mathbf{u} \sim N_p(\mathbf{m}, (\nu/w)\mathbf{S})$ . Write  $D = (\mathbf{u} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{u} - \mathbf{m})$ . Integrating the normal density (Theorem 3.2.2) against the  $\chi^2(\nu)$  density of  $W$ , and dropping factors free of  $\mathbf{u}$ ,

$$\begin{aligned} \int_0^\infty w^{p/2} e^{-wD/(2\nu)} w^{\nu/2-1} e^{-w/2} dw &= \int_0^\infty w^{(\nu+p)/2-1} e^{-w(1+D/\nu)/2} dw \\ &\propto (1 + D/\nu)^{-(\nu+p)/2}, \end{aligned}$$

by the substitution  $w' = w(1 + D/\nu)$ .

■

## 12.6.2 Credible sets and predictive distributions in the conjugate model

### Theorem 12.6.2 — Credible sets and posterior predictive distribution

Let the prior be  $\text{NIG}(\mathbf{m}_0, \mathbf{V}_0, a_0, b_0)$ , so that by Theorem 7.6.2 the posterior is  $\text{NIG}(\mathbf{m}_n, \mathbf{V}_n, a_n, b_n)$ . Put  $\nu = 2a_n$  and  $\tau^2 = b_n/a_n$ .

- (a) For  $\boldsymbol{\lambda} \neq \mathbf{0}$ , the interval

$$\boldsymbol{\lambda}^\top \mathbf{m}_n \pm t_{\nu, \alpha/2} \tau \sqrt{\boldsymbol{\lambda}^\top \mathbf{V}_n \boldsymbol{\lambda}}$$

is the  $1 - \alpha$  HPD credible interval for  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ ; it is also equal-tailed.

- (b) For a  $p \times q$  matrix  $\mathbf{A}$  of rank  $q$ , the set

$$\left\{ \boldsymbol{\phi}_0 : (\boldsymbol{\phi}_0 - \mathbf{A}^\top \mathbf{m}_n)^\top (\mathbf{A}^\top \mathbf{V}_n \mathbf{A})^{-1} (\boldsymbol{\phi}_0 - \mathbf{A}^\top \mathbf{m}_n) \leq q \tau^2 F_\alpha(q, \nu) \right\}$$

is the  $1 - \alpha$  HPD credible region for  $\mathbf{A}^\top \boldsymbol{\beta}$ .

- (c) Let  $\mathbf{Y}_0 = \mathbf{X}_0 \boldsymbol{\beta} + \boldsymbol{\varepsilon}_0$  be  $k$  future observations, with  $\boldsymbol{\varepsilon}_0 \sim \text{N}_k(\mathbf{0}, \sigma^2 \mathbf{I}_k)$  independent of  $\mathbf{Y}$  and of  $\boldsymbol{\beta}$  given  $\sigma^2$ . The posterior predictive distribution of  $\mathbf{Y}_0$  is

$$\mathbf{Y}_0 \mid \mathbf{y} \sim t_\nu(\mathbf{X}_0 \mathbf{m}_n, \tau^2(\mathbf{I}_k + \mathbf{X}_0 \mathbf{V}_n \mathbf{X}_0^\top)).$$

In particular, for one future observation at  $\mathbf{x}_0$ , the interval  $\mathbf{x}_0^\top \mathbf{m}_n \pm t_{\nu, \alpha/2} \tau \sqrt{1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0}$  has posterior predictive probability  $1 - \alpha$ .

- (d)  $[2b_n/\chi_{\alpha/2}^2(\nu), 2b_n/\chi_{1-\alpha/2}^2(\nu)]$  is an equal-tailed  $1 - \alpha$  credible interval for  $\sigma^2$ .

*Proof.* By Theorem 7.6.2 and Lemma 7.6.1,  $\boldsymbol{\beta} \mid \mathbf{y} \sim t_\nu(\mathbf{m}_n, \tau^2 \mathbf{V}_n)$  and  $2b_n/\sigma^2 \mid \mathbf{y} \sim \chi^2(\nu)$ .

- (a) By Lemma 12.6.1(a) with  $\mathbf{A} = \boldsymbol{\lambda}^\top$ ,  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \boldsymbol{\lambda}^\top \mathbf{m}_n + \tau \sqrt{\boldsymbol{\lambda}^\top \mathbf{V}_n \boldsymbol{\lambda}} T$  with  $T \sim t(\nu)$ . The  $t(\nu)$  density is symmetric and decreasing in  $|x|$ , so the symmetric interval is both equal-tailed and HPD.
- (b) By Lemma 12.6.1(a),  $\mathbf{A}^\top \boldsymbol{\beta} \mid \mathbf{y} \sim t_\nu(\mathbf{A}^\top \mathbf{m}_n, \tau^2 \mathbf{A}^\top \mathbf{V}_n \mathbf{A})$ . By (b) of the lemma the quadratic form divided by  $q\tau^2$  is  $F(q, \nu)$ , so the set has posterior probability  $1 - \alpha$ . By (c) the density is a decreasing function of the quadratic form, so the set has the form  $\{\text{density} \geq k\}$ .
- (c) Given  $\sigma^2$  and  $\mathbf{y}$ ,  $\boldsymbol{\beta} \sim \text{N}_p(\mathbf{m}_n, \sigma^2 \mathbf{V}_n)$  and  $\boldsymbol{\varepsilon}_0 \sim \text{N}_k(\mathbf{0}, \sigma^2 \mathbf{I}_k)$  independently, so  $\mathbf{Y}_0 \sim \text{N}_k(\mathbf{X}_0 \mathbf{m}_n, \sigma^2(\mathbf{X}_0 \mathbf{V}_n \mathbf{X}_0^\top + \mathbf{I}_k))$  (Theorem 3.3.1). Together with  $\sigma^2 \mid \mathbf{y} \sim \text{IG}(a_n, b_n)$ , this says that  $(\mathbf{Y}_0, \sigma^2) \mid \mathbf{y}$  is  $\text{NIG}(\mathbf{X}_0 \mathbf{m}_n, \mathbf{I}_k + \mathbf{X}_0 \mathbf{V}_n \mathbf{X}_0^\top, a_n, b_n)$  (Definition 7.6.1),

the matrix being positive definite. Lemma 7.6.1 gives the  $t_\nu$  law, and (a) of the present theorem, applied to it with  $k = 1$ , gives the interval.

(d)  $\Pr\{\chi_{1-\alpha/2}^2(\nu) \leq 2b_n/\sigma^2 \leq \chi_{\alpha/2}^2(\nu) \mid \mathbf{y}\} = 1 - \alpha$ ; invert the inequalities.

■

The formulas mirror the frequentist ones:  $\hat{\boldsymbol{\beta}}$  becomes  $\mathbf{m}_n$ ,  $\mathbf{G}$  becomes  $\mathbf{V}_n = (\mathbf{V}_0^{-1} + \mathbf{X}^\top \mathbf{X})^{-1}$ ,  $s^2$  becomes  $\tau^2$ , and  $n - r$  becomes  $\nu = 2a_0 + n$ , which counts every observation plus  $2a_0$  “prior observations” about  $\sigma^2$ . The 1 in  $1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0$  is again the new error.

### Corollary 12.6.3 — The flat prior reproduces the frequentist sets

Let  $\text{rank}(\mathbf{X}) = p < n$ ,  $\text{SSE} > 0$ , and let the prior be  $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$ . Then the credible interval (a), the region (b), the predictive interval (c) and the interval for  $\sigma^2$  (d) of Theorem 12.6.2 are, respectively, the  $t$  interval of Theorem 12.1.1, the confidence ellipsoid of Theorem 12.2.1, the prediction interval of Theorem 12.3.1 and the interval of Proposition 12.1.4.

*Proof.* By Corollary 7.6.4 the posterior is  $\text{NIG}(\hat{\boldsymbol{\beta}}, (\mathbf{X}^\top \mathbf{X})^{-1}, (n - p)/2, \text{SSE}/2)$ . The proof of Theorem 12.6.2 used only that the posterior is normal-inverse-gamma, so its conclusions hold with  $\mathbf{m}_n = \hat{\boldsymbol{\beta}}$ ,  $\mathbf{V}_n = (\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{G}$ ,  $\nu = n - p = n - r$ ,  $\tau^2 = \text{SSE}/(n - p) = s^2$  and  $2b_n = \text{SSE}$ . Substituting gives the four frequentist sets.

■

The same set then carries two readings: “this procedure covers the fixed truth in 95% of samples” and “given these data, the parameter lies here with probability 0.95”. The agreement is special; it fails for proper priors and for most other models.

## 12.6.3 A worked example

### Example. (Predicting a day of stack loss)

Take the stack loss regression with the engineer’s prior of Example (A prior for the stack loss plant), and consider a new day with air flow 65, water temperature 23 and acid concentration 87. The posterior has  $a_n = 13.5$ , so  $\nu = 27$  and  $t_{27,0.025} = 2.052$ . By Theorem 12.6.2, the 95% credible interval for the mean stack loss at these conditions is  $23.12 \pm 1.51$ , and the 95% posterior predictive interval for the day’s stack loss is  $23.12 \pm 6.11$ . The least squares answers are  $23.15 \pm 1.77$  for the mean and  $23.15 \pm 7.07$  for the new day, with  $t_{17,0.025} = 2.110$ . The Bayesian intervals are shorter because the prior informs the slopes, shrinking  $\mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0$ , and  $\sigma^2$ , raising the degrees of freedom from 17 to 27.

The 2.5% and 97.5% points of 400000 posterior predictive draws are 17.00 and 29.21, matching the  $t$  interval within Monte Carlo error. The simulated posterior probability of the credible ellipse of (b) for the air-flow and water-temperature coefficients is 0.9503.

```

import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]]])
n, p = X.shape
XtX, Xty = X.T @ X, X.T @ y

def nig_posterior(m0, P0, a0, b0, y=y):
    """Conjugate update; P0 is the prior precision  $V_0^{-1}$  (per unit  $\sigma^2$ )."""
    Vn = np.linalg.inv(P0 + XtX)
    mn = Vn @ (P0 @ m0 + X.T @ y)
    an = a0 + n / 2
    bn = b0 + (y @ y + m0 @ P0 @ m0 - mn @ (P0 + XtX) @ mn) / 2
    return mn, Vn, an, bn

m0 = np.array([0.0, 1.0, 1.0, 0.0]) # the prior of Chapter 7
P0 = np.linalg.inv(np.diag([100.0**2, 0.1**2, 0.2**2, 0.1**2]))
a0, b0 = 3.0, 18.0
mn, Vn, an, bn = nig_posterior(m0, P0, a0, b0)

```

[Run this code online](#)

```

x0 = np.array([1.0, 65.0, 23.0, 87.0]) # a new operating condition
qb = stats.t.ppf(0.975, 2 * an)
centre = x0 @ mn
half_mean = qb * np.sqrt(bn / an * (x0 @ Vn @ x0)) # credible interval for  $x_0^T$ 
beta
half_new = qb * np.sqrt(bn / an * (1 + x0 @ Vn @ x0)) # posterior predictive
interval for  $Y_0$ 

beta_hat = np.linalg.solve(XtX, Xty) # frequentist intervals, for
comparison
s2 = np.sum((y - X @ beta_hat) ** 2) / (n - p)
qf = stats.t.ppf(0.975, n - p)
h0 = x0 @ np.linalg.solve(XtX, x0)
print(f"Bayes: mean response {centre:.2f} +- {half_mean:.2f}; new day {centre:.2f}
+- {half_new:.2f}")
print(f"least squares: mean response {x0 @ beta_hat:.2f} +- {qf * np.sqrt(s2 * h0)
:.2f}; "
f"new day +- {qf * np.sqrt(s2 * (1 + h0)):.2f}")

```

[Run this code online](#)

## 12.6.4 How often does a credible interval cover?

One coverage statement holds for every proper prior.

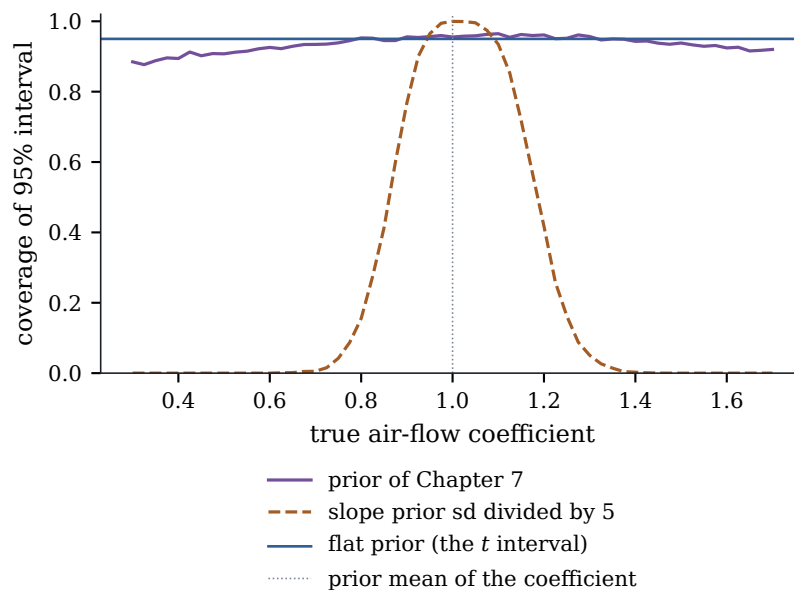


Figure 12.6.1: Frequentist coverage of the 95% credible interval for the air-flow coefficient against the true coefficient, for the prior of Chapter 7 and for a prior with slope standard deviations divided by five (4000 data sets per point). The flat-prior interval covers 0.95 everywhere.

### Proposition 12.6.4 — Coverage averaged over the prior

Let the prior  $\pi$  be proper, and let  $C(\mathbf{y})$  be a  $1 - \alpha$  credible set for  $\theta$  for every  $\mathbf{y}$ . If  $(\beta, \sigma^2)$  is drawn from  $\pi$  and then  $\mathbf{Y}$  from the model,  $\Pr\{\theta \in C(\mathbf{Y})\} \geq 1 - \alpha$ , with equality if every  $C(\mathbf{y})$  has posterior probability exactly  $1 - \alpha$ . Equivalently, the frequentist coverage  $\Pr_{\beta, \sigma^2}\{\theta \in C(\mathbf{Y})\}$ , averaged over  $\pi$ , is at least  $1 - \alpha$ .

*Proof.* Under the joint distribution of parameters and data, the conditional distribution of the parameters given  $\mathbf{Y} = \mathbf{y}$  is the posterior. By the tower property,  $\Pr\{\theta \in C(\mathbf{Y})\} = \mathbb{E}[\Pr\{\theta \in C(\mathbf{Y}) \mid \mathbf{Y}\}] \geq 1 - \alpha$ . Conditioning instead on the parameters gives the second form. ■

The average hides variation: coverage is high where the prior is right and low where it is wrong. Consider the credible interval for the air-flow coefficient in [Example \(Predicting a day of stack loss\)](#), and simulate its frequentist coverage when the true air-flow coefficient is varied, the other coefficients are held at their least squares values, and  $\sigma^2 = s^2$ . [Figure 12.6.1](#) shows the result. With the prior of Chapter 7, the coverage is 0.959 when the truth is the prior mean 1, and 0.936 at 0.7 and at 1.5. It ranges from 0.877 to 0.965 over the plotted range. This prior is mild, and prior–data conflict inflates  $b_n$ , which widens the interval. A prior with slope standard deviations five times smaller behaves very differently: its coverage is 1.000 at the prior mean but 0.007 at 0.7, close to the least squares estimate, and 0.000 at 1.5. Averaged over the prior of Chapter 7, the simulated coverage is 0.9512 (40000 draws), as [Proposition 12.6.4](#) requires. The flat-prior interval is the  $t$  interval and covers exactly 95% at every parameter value.

An informative prior makes intervals shorter, and better when the prior is sound; when it is not, they are confidently wrong. The frequentist interval makes no bet and pays in length. [Chapter 27](#) meets the same trade-off as bias against variance.

## 12.6.5 Exercises

### 12.6.5.1 A. Check your understanding

#### Exercise A1.

Is the equal-tailed interval of [Theorem 12.6.2\(d\)](#) the HPD interval for  $\sigma^2$ ? Describe the HPD interval and explain the difference.

*Solution.* No. The inverse gamma density is skewed to the right, so the HPD interval  $[l, u]$ , with  $\pi(l | \mathbf{y}) = \pi(u | \mathbf{y})$ , is shifted toward smaller values. It also depends on the parameterization: the HPD interval for  $\log \sigma^2$  does not transform into the one for  $\sigma^2$ , while equal-tailed intervals do.

#### Exercise A2.

For  $\nu > 2$ , find the posterior predictive mean and variance of a single future observation at  $\mathbf{x}_0$ . Why is the variance larger than  $\tau^2(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0)$ ?

*Solution.* From [Theorem 12.6.2\(c\)](#),  $Y_0 | \mathbf{y} = \mathbf{x}_0^\top \mathbf{m}_n + \tau \sqrt{1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0} T$  with  $T \sim t(\nu)$ , so the mean is  $\mathbf{x}_0^\top \mathbf{m}_n$  and the variance is  $\tau^2(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0) \nu/(\nu - 2)$ . The extra factor comes from the uncertainty about  $\sigma^2$ : given  $\sigma^2$  the predictive variance is  $\sigma^2(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0)$ , and averaging over the posterior gives  $E(\sigma^2 | \mathbf{y}) = b_n/(a_n - 1)$ , which exceeds  $\tau^2 = b_n/a_n$  by the factor  $a_n/(a_n - 1) = \nu/(\nu - 2)$ .

### 12.6.5.2 B. Practice

#### Exercise B1.

Under the  $g$ -prior of Proposition 7.6.3(c) with  $\mathbf{m}_0 = \mathbf{0}$ , show that the credible interval for  $\lambda^\top \boldsymbol{\beta}$  is centred at  $\{g/(1+g)\} \lambda^\top \hat{\boldsymbol{\beta}}$  and has half-width  $t_{\nu, \alpha/2} \tau \sqrt{g/(1+g)} \sqrt{\lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda}$ . Compare it with the  $t$  interval as  $g \rightarrow \infty$ , paying attention to  $\nu$  and  $\tau^2$ .

*Solution.* With  $\mathbf{V}_0 = g(\mathbf{X}^\top \mathbf{X})^{-1}$ ,  $\mathbf{V}_n = \{g/(1+g)\}(\mathbf{X}^\top \mathbf{X})^{-1}$  and  $\mathbf{m}_n = \{g/(1+g)\} \hat{\boldsymbol{\beta}}$  (Proposition 7.6.3(c)). Substitute in [Theorem 12.6.2\(a\)](#). As  $g \rightarrow \infty$  the centre and the matrix tend to those of least squares, but  $\nu = 2a_0 + n$  stays larger than  $n - p$  and  $\tau^2 = b_n/a_n$  tends to  $(2b_0 + \text{SSE})/(2a_0 + n)$ , not to  $s^2$ . The interval does not become the  $t$  interval: that needs the flat prior on  $\sigma^2$  as well, with  $a_0 = -p/2$  and  $b_0 = 0$  (Corollary 7.6.4).

#### Exercise B2.

Two future observations are taken at the same  $\mathbf{x}_0$ . Show that, under the posterior predictive distribution, they are correlated, and find the correlation. Why are they correlated when their errors are independent? Are they independent under the frequentist model?

*Solution.* By [Theorem 12.6.2\(c\)](#) with  $\mathbf{X}_0$  having two equal rows  $\mathbf{x}_0^\top$ , the scale matrix is  $\tau^2(\mathbf{I}_2 + h\mathbf{1}\mathbf{1}^\top)$  with  $h = \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0$ , so the correlation is  $h/(1+h)$ . Both observations share the unknown  $\boldsymbol{\beta}$  (and  $\sigma^2$ ), and uncertainty about the shared mean correlates them. Under the frequentist model with fixed parameters the two observations are independent, but the two prediction errors  $Y_{0j} - \hat{Y}_0$  share  $\hat{Y}_0$  and have the same correlation  $h_0/(1+h_0)$  under the flat prior.

### 12.6.5.3 C. Going deeper

#### Exercise C1.

Let  $\mathbf{X}$  have rank  $r < p$  and take the flat prior  $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$ . Show that the posterior of  $\boldsymbol{\beta}$  is improper, but that for estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  the posterior of  $(\boldsymbol{\lambda}^\top \boldsymbol{\beta}, \sigma^2)$  can be defined by reparameterizing with  $\boldsymbol{\gamma} = \mathbf{U}^\top \boldsymbol{\beta}$  and  $\boldsymbol{\delta} = \mathbf{N}^\top \boldsymbol{\beta}$ , where the columns of  $\mathbf{U}$  and  $\mathbf{N}$  are orthonormal bases of  $\mathcal{C}(\mathbf{X}^\top)$  and  $\mathcal{N}(\mathbf{X})$ , and integrating over nothing but  $\boldsymbol{\gamma}$ . Show that the resulting credible interval is the  $t$  interval of Theorem 12.1.1 with  $n - r$  degrees of freedom, provided the flat prior is taken as  $1/\sigma^2$  in  $(\boldsymbol{\gamma}, \sigma^2)$  alone.

*Solution.* The likelihood depends on  $\boldsymbol{\beta}$  only through  $\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\mathbf{U}\boldsymbol{\gamma}$ , since  $\mathbf{X}\mathbf{N} = \mathbf{0}$ . So the posterior is flat in  $\boldsymbol{\delta}$ , which is not integrable: the posterior of  $\boldsymbol{\beta}$  is improper. In the model  $\mathbf{Y} \sim N_n(\mathbf{X}\mathbf{U}\boldsymbol{\gamma}, \sigma^2\mathbf{I})$  the matrix  $\mathbf{X}\mathbf{U}$  has full column rank  $r$ , and Corollary 7.6.4 with  $p$  replaced by  $r$  gives a proper posterior with  $t(n - r)$  marginals. An estimable  $\boldsymbol{\lambda}$  lies in  $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{U})$ , so  $\boldsymbol{\lambda} = \mathbf{U}\mathbf{U}^\top \boldsymbol{\lambda}$  and  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = (\mathbf{U}^\top \boldsymbol{\lambda})^\top \boldsymbol{\gamma}$  involves  $\boldsymbol{\gamma}$  only. Its credible interval is centred at  $(\mathbf{U}^\top \boldsymbol{\lambda})^\top \hat{\boldsymbol{\gamma}} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  with scale  $s^2(\mathbf{U}^\top \boldsymbol{\lambda})^\top (\mathbf{U}^\top \mathbf{X}^\top \mathbf{X} \mathbf{U})^{-1} (\mathbf{U}^\top \boldsymbol{\lambda})$ , and  $\mathbf{U}(\mathbf{U}^\top \mathbf{X}^\top \mathbf{X} \mathbf{U})^{-1} \mathbf{U}^\top$  is a generalized inverse of  $\mathbf{X}^\top \mathbf{X}$  (check  $\mathbf{X}^\top \mathbf{X} \mathbf{U}(\mathbf{U}^\top \mathbf{X}^\top \mathbf{X} \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{X}^\top \mathbf{X} = \mathbf{X}^\top \mathbf{X}$  using  $\mathbf{X} = \mathbf{X}\mathbf{U}\mathbf{U}^\top$ ), so the scale is  $s^2 \boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}$ .

#### Exercise C2.

In the calibration model of [Section 12.5](#), give  $x_0$  a normal prior independent of the regression parameters, and use the flat prior  $1/\sigma^2$  for  $(\beta_0, \beta_1, \sigma^2)$ . Write down the posterior density of  $x_0$  up to a constant, and show that it is proper even when the slope is not significant. Compare the behaviour of the resulting credible interval with the Fieller set in the weak assay of Example (Calibrating an assay).